Cloud Computing (CS3081)

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Github: https://github.com/Abhishek-singh88/cloud-assignment

1. Fundamental Learning & Conceptual Takeaways

Undertaking this mini project provided me with an end to end practical understanding of the intersection between ML systems and cloud infrastructure. Rather than treating model training and service deployment as isolated tasks, this project demonstrated that serving AI models reliably at scale is fundamentally an exercise in distributed systems constraints, memory management and ~~sc~~ resilience engineering.

(a) Sequence level knowledge distillation on consumer hardware.

(b) Cloud Native storage Decoupling & containers Immutability.

(c) Systems Automation & stateless Network security.

2. Technical challenges Encountered & Engineering solutions.

Challenge 1: Ollama cross-container Network Refusal in k3d cluster.

Solution: I reconfigured and launched the host Ollama service with OLLAMA_HOST = 0.0.0:11434, enabled OLLAMA_FLASH_ATTENTION = true, and set OLLAMA_NUM_PARALLEL = 2, enabling multiple worker pods to query concurrent inference slots without starvation.

Challenge 2: SeaweedFS container OOM Termination during model uploads

Solution: I updated k8s/data/seaweedfs.yaml to provision a 2 GiB persistentVolumeclaim (seaweedfs.pvc) mounted under /data and raised the memory limit to 1.5 GiB, permanently resolving storage persistence and memory starvation.

3. Empirical Findings, SRE Observations & What could be done better

Key Empirical Findings & SRE observations

1. Chaos Recovery & SRE Observations: Deleting an active pod under load resulted in recovery within 18.93s with zero dropped connections. Traefik Ingress evenly distributed traffic across replicas (53.4 v. / 46.6 v.)

2. Economic serving cost: Under the project rate card, the dual replica deployment costs 7.9 INR/hr. Serving 2052 requests/hr result in a unit cost of 3.85 INR per 1000 request.

What could have been done better

1. Pre-hydrated persistent volumes: Using a Shared ReadWriteMany volume with safetensors memory mapping (mmap = True) would eliminate network weight downloads reducing cold starts from 18.9s to sub-seconds levels.

. Stateful eBPF security policies: Replacing stateless NACLs with cilium eBPF network policies would allow connection tracking and identity aware filtering without managing ephemeral port return rules.